In [1]:
```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns
from sklearn.model_selection import train_test_split
from sklearn.preprocessing import StandardScaler
from sklearn.metrics import classification_report, confusion_matrix, ConfusionMatrixDisplay, accuracy_score, pr
import tensorflow as tf
from tensorflow.keras.models import Sequential
from tensorflow.keras.layers import Dense

%matplotlib inline
```

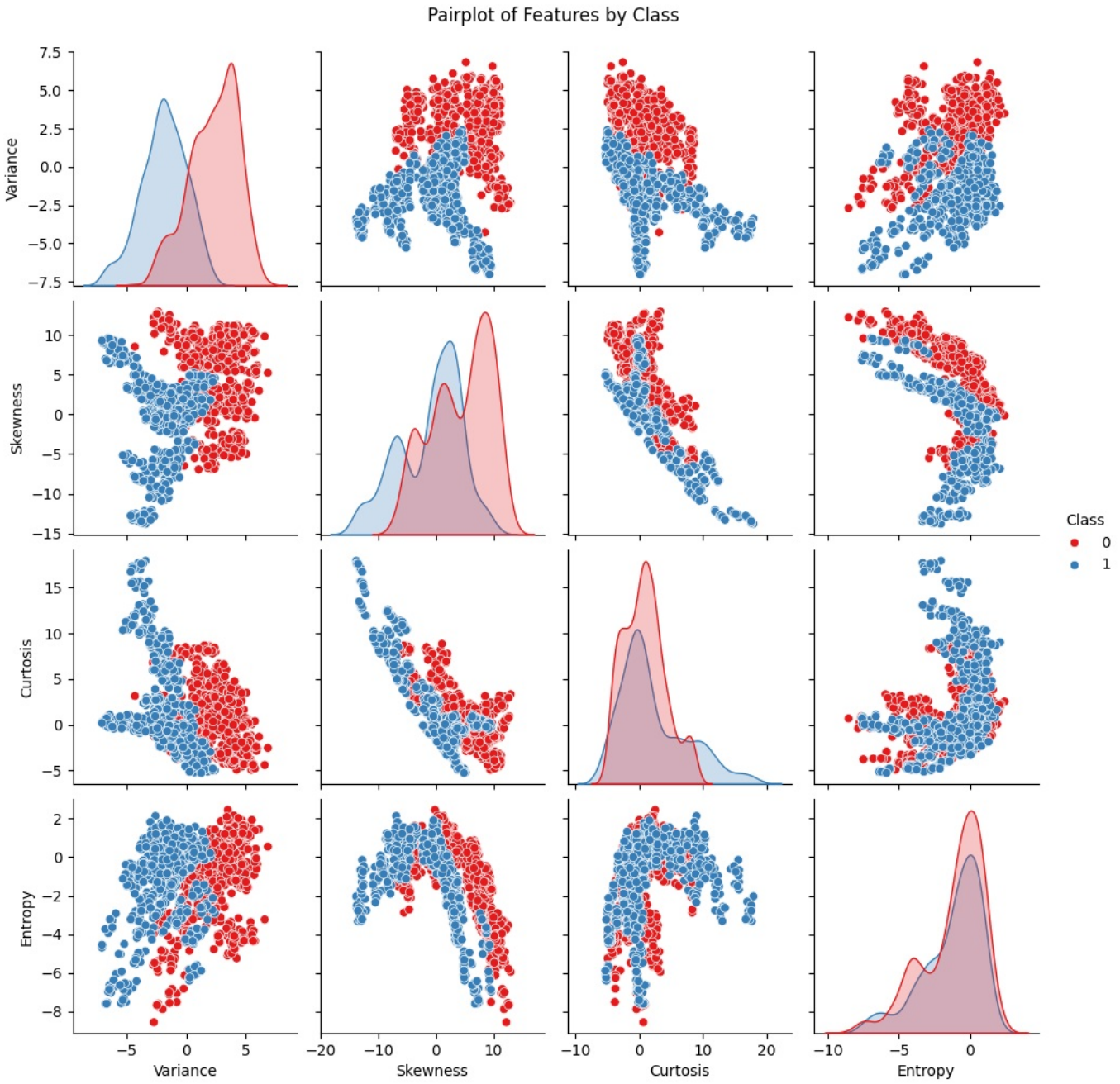
In [2]:
```
# Load dataset
df = pd.read_csv('bill_authentication.csv')
print(f"Dataset shape: {df.shape}")
display(df.head())
```

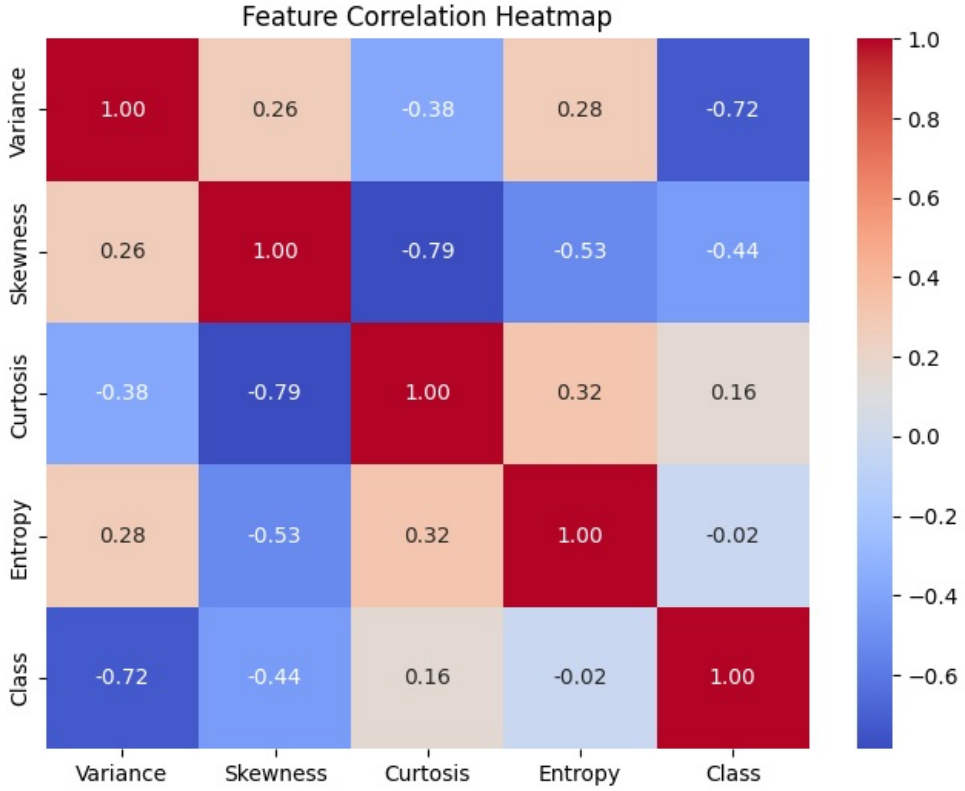
```
Dataset shape: (1372, 5)
```

	Variance	Skewness	Curtosis	Entropy	Class
0	3.62160	8.6661	-2.8073	-0.44699	0
1	4.54590	8.1674	-2.4586	-1.46210	0
2	3.86600	-2.6383	1.9242	0.10645	0
3	3.45660	9.5228	-4.0112	-3.59440	0
4	0.32924	-4.4552	4.5718	-0.98880	0

In [3]:
```
# Exploratory Data Analysis: Pairplot
sns.pairplot(df, hue='Class', palette='Set1')
plt.suptitle('Pairplot of Features by Class', y=1.02)
plt.show()

# Correlation Heatmap
plt.figure(figsize=(8, 6))
sns.heatmap(df.corr(), annot=True, cmap='coolwarm', fmt='.2f')
plt.title('Feature Correlation Heatmap')
plt.show()
```


Pairplot of Features by Class
Variance
Skewness
Curtosis
Entropy
Class
0
1


Feature Correlation Heatmap
Variance
Skewness
Curtosis
Entropy
Class
1.00 0.26 -0.38 0.28 -0.72
0.26 1.00 -0.79 -0.53 -0.44
-0.38 -0.79 1.00 0.32 0.16
0.28 -0.53 0.32 1.00 -0.02
-0.72 -0.44 0.16 -0.02 1.00

In [4]:
```
# Prepare features and labels
X = df.drop('Class', axis=1)
y = df['Class']

# Split data (80% train, 20% test)
X_train, X_test, y_train, y_test = train_test_split(X, y, test_size=0.20, random_state=42)

# Feature Scaling (Crucial for Neural Networks)
scaler = StandardScaler()
X_train_scaled = scaler.fit_transform(X_train)
X_test_scaled = scaler.transform(X_test)
```

In [5]:
```
# Define ANN Model
model = Sequential([
    Dense(16, activation='relu', input_shape=(X_train_scaled.shape[1],)),
    Dense(8, activation='relu'),
    Dense(1, activation='sigmoid') # Binary classification
])

# Compile Model
model.compile(optimizer='adam', loss='binary_crossentropy', metrics=['accuracy'])

model.summary()
```

```
e:\anaconda3\envs\tf_env\lib\site-packages\keras\src\layers\core\dense.py:95: UserWarning: Do not pass an `input
_shape`/`input_dim` argument to a layer. When using Sequential models, prefer using an `Input(shape)` object as
the first layer in the model instead.
  super().__init__(activity_regularizer=activity_regularizer, **kwargs)
```

Model: "sequential"

Layer (type)	Output Shape	Param #
dense (Dense)	(None, 16)	80
dense_1 (Dense)	(None, 8)	136
dense_2 (Dense)	(None, 1)	9

Total params: 225 (900.00 B)

Trainable params: 225 (900.00 B)

Non-trainable params: 0 (0.00 B)

In [6]:
```
# Train Model
history = model.fit(
    X_train_scaled, y_train,
    epochs=50,
    batch_size=32,
    validation_split=0.2,
    verbose=1
)
```

```
Epoch 1/50
28/28 ━━━━━━━━━━━━━━━━━━━━ 1s 9ms/step - accuracy: 0.7013 - loss: 0.6283 - val_accuracy: 0.7636 - val_loss: 0.59
75
Epoch 2/50
28/28 ━━━━━━━━━━━━━━━━━━━━ 0s 4ms/step - accuracy: 0.7548 - loss: 0.5766 - val_accuracy: 0.7818 - val_loss: 0.55
27
Epoch 3/50
28/28 ━━━━━━━━━━━━━━━━━━━━ 0s 4ms/step - accuracy: 0.7913 - loss: 0.5293 - val_accuracy: 0.8000 - val_loss: 0.50
74
Epoch 4/50
28/28 ━━━━━━━━━━━━━━━━━━━━ 0s 4ms/step - accuracy: 0.8107 - loss: 0.4828 - val_accuracy: 0.8273 - val_loss: 0.46
03
Epoch 5/50
28/28 ━━━━━━━━━━━━━━━━━━━━ 0s 3ms/step - accuracy: 0.8335 - loss: 0.4351 - val_accuracy: 0.8500 - val_loss: 0.41
07
Epoch 6/50
28/28 ━━━━━━━━━━━━━━━━━━━━ 0s 3ms/step - accuracy: 0.8461 - loss: 0.3858 - val_accuracy: 0.8727 - val_loss: 0.35
75
Epoch 7/50
28/28 ━━━━━━━━━━━━━━━━━━━━ 0s 3ms/step - accuracy: 0.8746 - loss: 0.3316 - val_accuracy: 0.9091 - val_loss: 0.29
86
Epoch 8/50
28/28 ━━━━━━━━━━━━━━━━━━━━ 0s 3ms/step - accuracy: 0.9054 - loss: 0.2740 - val_accuracy: 0.9227 - val_loss: 0.23
95
Epoch 9/50
28/28 ━━━━━━━━━━━━━━━━━━━━ 0s 3ms/step - accuracy: 0.9270 - loss: 0.2141 - val_accuracy: 0.9409 - val_loss: 0.18
39
Epoch 10/50
28/28 ━━━━━━━━━━━━━━━━━━━━ 0s 3ms/step - accuracy: 0.9453 - loss: 0.1674 - val_accuracy: 0.9591 - val_loss: 0.14
24
```

```
Epoch 11/50
28/28 ━━━━━━━━━━━━━━━━━━━━ 0s 3ms/step - accuracy: 0.9635 - loss: 0.1319 - val_accuracy: 0.9773 - val_loss: 0.11
32
Epoch 12/50
28/28 ━━━━━━━━━━━━━━━━━━━━ 0s 4ms/step - accuracy: 0.9726 - loss: 0.1076 - val_accuracy: 0.9818 - val_loss: 0.09
25
Epoch 13/50
28/28 ━━━━━━━━━━━━━━━━━━━━ 0s 4ms/step - accuracy: 0.9761 - loss: 0.0906 - val_accuracy: 0.9818 - val_loss: 0.07
73
Epoch 14/50
28/28 ━━━━━━━━━━━━━━━━━━━━ 0s 3ms/step - accuracy: 0.9783 - loss: 0.0779 - val_accuracy: 0.9818 - val_loss: 0.06
63
Epoch 15/50
28/28 ━━━━━━━━━━━━━━━━━━━━ 0s 3ms/step - accuracy: 0.9806 - loss: 0.0677 - val_accuracy: 0.9818 - val_loss: 0.05
73
Epoch 16/50
28/28 ━━━━━━━━━━━━━━━━━━━━ 0s 3ms/step - accuracy: 0.9829 - loss: 0.0601 - val_accuracy: 0.9864 - val_loss: 0.04
97
Epoch 17/50
28/28 ━━━━━━━━━━━━━━━━━━━━ 0s 3ms/step - accuracy: 0.9840 - loss: 0.0532 - val_accuracy: 0.9864 - val_loss: 0.04
37
Epoch 18/50
28/28 ━━━━━━━━━━━━━━━━━━━━ 0s 3ms/step - accuracy: 0.9863 - loss: 0.0479 - val_accuracy: 0.9864 - val_loss: 0.03
89
Epoch 19/50
28/28 ━━━━━━━━━━━━━━━━━━━━ 0s 4ms/step - accuracy: 0.9886 - loss: 0.0432 - val_accuracy: 0.9955 - val_loss: 0.03
48
Epoch 20/50
28/28 ━━━━━━━━━━━━━━━━━━━━ 0s 3ms/step - accuracy: 0.9932 - loss: 0.0393 - val_accuracy: 0.9955 - val_loss: 0.03
13
Epoch 21/50
28/28 ━━━━━━━━━━━━━━━━━━━━ 0s 4ms/step - accuracy: 0.9932 - loss: 0.0359 - val_accuracy: 0.9955 - val_loss: 0.02
83
Epoch 22/50
28/28 ━━━━━━━━━━━━━━━━━━━━ 0s 3ms/step - accuracy: 0.9932 - loss: 0.0329 - val_accuracy: 0.9955 - val_loss: 0.02
57
Epoch 23/50
28/28 ━━━━━━━━━━━━━━━━━━━━ 0s 3ms/step - accuracy: 0.9932 - loss: 0.0306 - val_accuracy: 0.9955 - val_loss: 0.02
34
Epoch 24/50
28/28 ━━━━━━━━━━━━━━━━━━━━ 0s 3ms/step - accuracy: 0.9932 - loss: 0.0279 - val_accuracy: 0.9955 - val_loss: 0.02
11
Epoch 25/50
28/28 ━━━━━━━━━━━━━━━━━━━━ 0s 3ms/step - accuracy: 0.9932 - loss: 0.0258 - val_accuracy: 0.9955 - val_loss: 0.01
93
Epoch 26/50
28/28 ━━━━━━━━━━━━━━━━━━━━ 0s 3ms/step - accuracy: 1.0000 - loss: 0.0235 - val_accuracy: 1.0000 - val_loss: 0.01
74
Epoch 27/50
28/28 ━━━━━━━━━━━━━━━━━━━━ 0s 3ms/step - accuracy: 1.0000 - loss: 0.0216 - val_accuracy: 1.0000 - val_loss: 0.01
58
Epoch 28/50
28/28 ━━━━━━━━━━━━━━━━━━━━ 0s 4ms/step - accuracy: 1.0000 - loss: 0.0199 - val_accuracy: 1.0000 - val_loss: 0.01
46
Epoch 29/50
28/28 ━━━━━━━━━━━━━━━━━━━━ 0s 3ms/step - accuracy: 1.0000 - loss: 0.0185 - val_accuracy: 1.0000 - val_loss: 0.01
33
Epoch 30/50
28/28 ━━━━━━━━━━━━━━━━━━━━ 0s 3ms/step - accuracy: 1.0000 - loss: 0.0171 - val_accuracy: 1.0000 - val_loss: 0.01
22
Epoch 31/50
28/28 ━━━━━━━━━━━━━━━━━━━━ 0s 3ms/step - accuracy: 1.0000 - loss: 0.0158 - val_accuracy: 1.0000 - val_loss: 0.01
10
Epoch 32/50
28/28 ━━━━━━━━━━━━━━━━━━━━ 0s 3ms/step - accuracy: 1.0000 - loss: 0.0145 - val_accuracy: 1.0000 - val_loss: 0.01
01
Epoch 33/50
28/28 ━━━━━━━━━━━━━━━━━━━━ 0s 3ms/step - accuracy: 1.0000 - loss: 0.0134 - val_accuracy: 1.0000 - val_loss: 0.00
93
Epoch 34/50
28/28 ━━━━━━━━━━━━━━━━━━━━ 0s 3ms/step - accuracy: 1.0000 - loss: 0.0126 - val_accuracy: 1.0000 - val_loss: 0.00
86
Epoch 35/50
28/28 ━━━━━━━━━━━━━━━━━━━━ 0s 3ms/step - accuracy: 1.0000 - loss: 0.0118 - val_accuracy: 1.0000 - val_loss: 0.00
80
Epoch 36/50
28/28 ━━━━━━━━━━━━━━━━━━━━ 0s 3ms/step - accuracy: 1.0000 - loss: 0.0110 - val_accuracy: 1.0000 - val_loss: 0.00
75
Epoch 37/50
28/28 ━━━━━━━━━━━━━━━━━━━━ 0s 3ms/step - accuracy: 1.0000 - loss: 0.0104 - val_accuracy: 1.0000 - val_loss: 0.00
70
Epoch 38/50
28/28 ━━━━━━━━━━━━━━━━━━━━ 0s 3ms/step - accuracy: 1.0000 - loss: 0.0099 - val_accuracy: 1.0000 - val_loss: 0.00
```

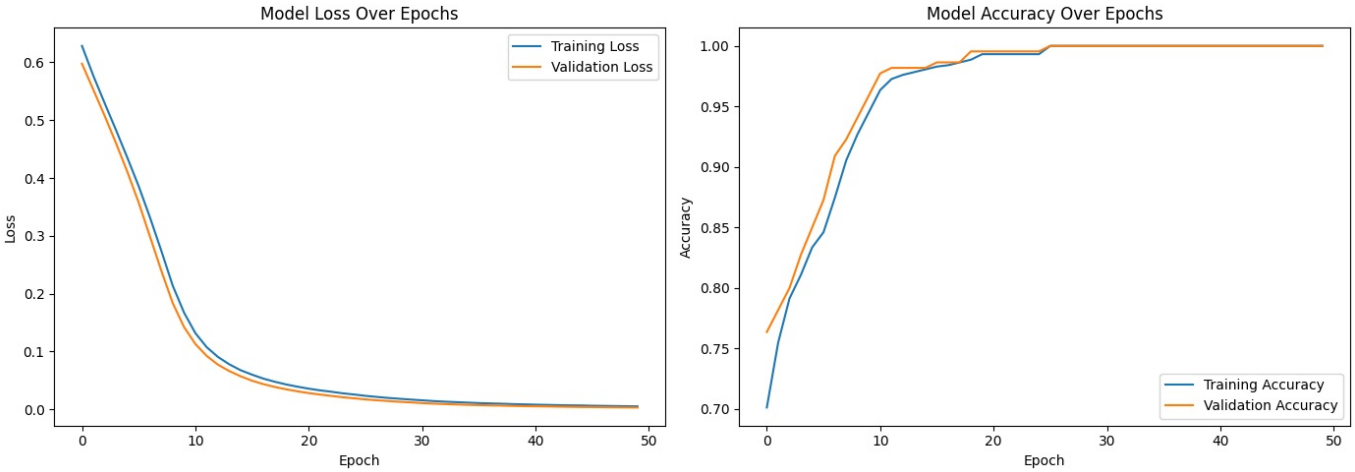
```
66
Epoch 39/50
28/28 ━━━━━━━━━━━━━━━━━━━━ 0s 3ms/step - accuracy: 1.0000 - loss: 0.0092 - val_accuracy: 1.0000 - val_loss: 0.00
61
Epoch 40/50
28/28 ━━━━━━━━━━━━━━━━━━━━ 0s 3ms/step - accuracy: 1.0000 - loss: 0.0088 - val_accuracy: 1.0000 - val_loss: 0.00
57
Epoch 41/50
28/28 ━━━━━━━━━━━━━━━━━━━━ 0s 3ms/step - accuracy: 1.0000 - loss: 0.0082 - val_accuracy: 1.0000 - val_loss: 0.00
54
Epoch 42/50
28/28 ━━━━━━━━━━━━━━━━━━━━ 0s 3ms/step - accuracy: 1.0000 - loss: 0.0078 - val_accuracy: 1.0000 - val_loss: 0.00
51
Epoch 43/50
28/28 ━━━━━━━━━━━━━━━━━━━━ 0s 3ms/step - accuracy: 1.0000 - loss: 0.0074 - val_accuracy: 1.0000 - val_loss: 0.00
48
Epoch 44/50
28/28 ━━━━━━━━━━━━━━━━━━━━ 0s 3ms/step - accuracy: 1.0000 - loss: 0.0071 - val_accuracy: 1.0000 - val_loss: 0.00
45
Epoch 45/50
28/28 ━━━━━━━━━━━━━━━━━━━━ 0s 3ms/step - accuracy: 1.0000 - loss: 0.0069 - val_accuracy: 1.0000 - val_loss: 0.00
43
Epoch 46/50
28/28 ━━━━━━━━━━━━━━━━━━━━ 0s 3ms/step - accuracy: 1.0000 - loss: 0.0064 - val_accuracy: 1.0000 - val_loss: 0.00
41
Epoch 47/50
28/28 ━━━━━━━━━━━━━━━━━━━━ 0s 3ms/step - accuracy: 1.0000 - loss: 0.0061 - val_accuracy: 1.0000 - val_loss: 0.00
38
Epoch 48/50
28/28 ━━━━━━━━━━━━━━━━━━━━ 0s 3ms/step - accuracy: 1.0000 - loss: 0.0058 - val_accuracy: 1.0000 - val_loss: 0.00
37
Epoch 49/50
28/28 ━━━━━━━━━━━━━━━━━━━━ 0s 3ms/step - accuracy: 1.0000 - loss: 0.0056 - val_accuracy: 1.0000 - val_loss: 0.00
35
Epoch 50/50
28/28 ━━━━━━━━━━━━━━━━━━━━ 0s 3ms/step - accuracy: 1.0000 - loss: 0.0054 - val_accuracy: 1.0000 - val_loss: 0.00
33
```

In [7]:
```python
# Plot Training History
fig, (ax1, ax2) = plt.subplots(1, 2, figsize=(14, 5))

# Loss Curve
ax1.plot(history.history['loss'], label='Training Loss')
ax1.plot(history.history['val_loss'], label='Validation Loss')
ax1.set_title('Model Loss Over Epochs')
ax1.set_xlabel('Epoch')
ax1.set_ylabel('Loss')
ax1.legend()

# Accuracy Curve
ax2.plot(history.history['accuracy'], label='Training Accuracy')
ax2.plot(history.history['val_accuracy'], label='Validation Accuracy')
ax2.set_title('Model Accuracy Over Epochs')
ax2.set_xlabel('Epoch')
ax2.set_ylabel('Accuracy')
ax2.legend()

plt.tight_layout()
plt.show()
```



In [8]:
```python
# Predictions
y_pred_prob = model.predict(X_test_scaled)
y_pred = (y_pred_prob > 0.5).astype(int).flatten()
```

```
# Print Classification Report
print("Classification Report:")
print(classification_report(y_test, y_pred))

# Calculate Metrics
accuracy = accuracy_score(y_test, y_pred)
precision = precision_score(y_test, y_pred, average='macro')
recall = recall_score(y_test, y_pred, average='macro')
f1 = f1_score(y_test, y_pred, average='macro')

print(f"Accuracy:  {accuracy:.4f}")
print(f"Precision: {precision:.4f}")
print(f"Recall:    {recall:.4f}")
print(f"F1-Score:  {f1:.4f}")
```

```
9/9 ━━━━━━━━━━━━━━━━━━━━ 0s 5ms/step
Classification Report:
              precision    recall  f1-score   support

           0       1.00      1.00      1.00       148
           1       1.00      1.00      1.00       127

    accuracy                           1.00       275
   macro avg       1.00      1.00      1.00       275
weighted avg       1.00      1.00      1.00       275

Accuracy:  1.0000
Precision: 1.0000
Recall:    1.0000
F1-Score:  1.0000
```

In [9]:

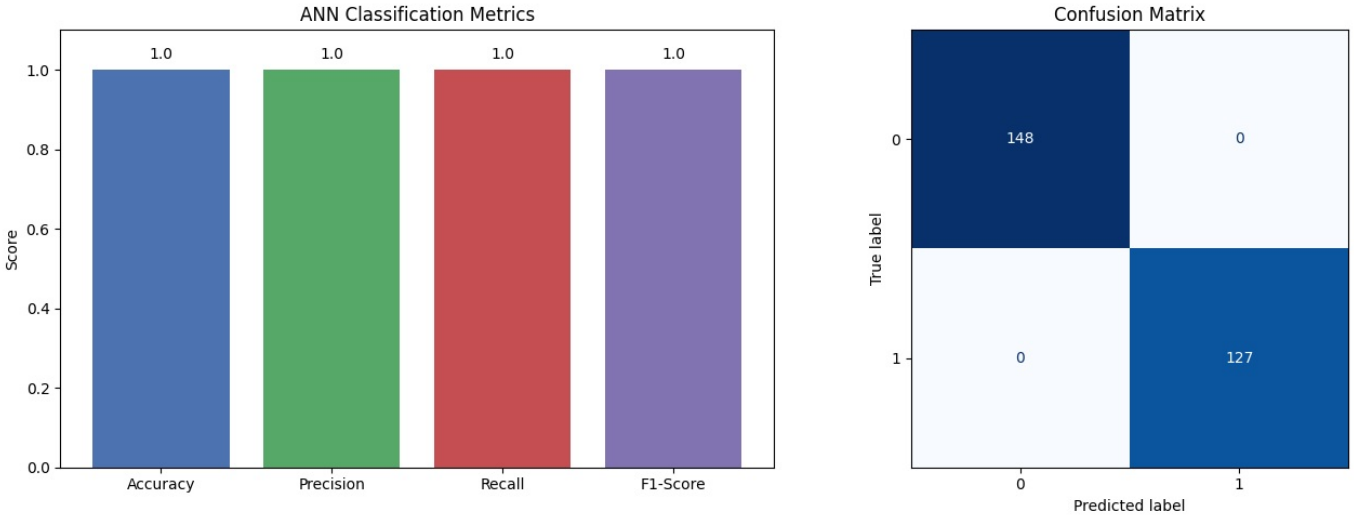
```
# Visualization of Metrics
fig, (ax1, ax2) = plt.subplots(1, 2, figsize=(14, 5))

# Plot 1: Classification Metrics Bar Chart
metrics = ['Accuracy', 'Precision', 'Recall', 'F1-Score']
values = [accuracy, precision, recall, f1]
bars = ax1.bar(metrics, values, color=['#4C72B0', '#55A868', '#C44E52', '#8172B2'])
ax1.set_ylim(0, 1.1)
ax1.set_title('ANN Classification Metrics')
ax1.set_ylabel('Score')

for bar in bars:
    yval = bar.get_height()
    ax1.text(bar.get_x() + bar.get_width()/2, yval + 0.02, round(yval, 4), ha='center', va='bottom')

# Plot 2: Confusion Matrix Heatmap
cm = confusion_matrix(y_test, y_pred)
disp = ConfusionMatrixDisplay(confusion_matrix=cm)
disp.plot(cmap='Blues', ax=ax2, colorbar=False)
ax2.set_title('Confusion Matrix')

plt.tight_layout()
plt.show()
```



In [10]:

```
# Plot ROC Curve
fpr, tpr, thresholds = roc_curve(y_test, y_pred_prob)
roc_auc = auc(fpr, tpr)

plt.figure(figsize=(8, 6))
plt.plot(fpr, tpr, color='darkorange', lw=2, label=f'ROC curve (area = {roc_auc:.4f})')
plt.plot([0, 1], [0, 1], color='navy', lw=2, linestyle='--')
plt.xlim([0.0, 1.0])
```

```
plt.ylim([0.0, 1.05])
plt.xlabel('False Positive Rate')
plt.ylabel('True Positive Rate')
plt.title('Receiver Operating Characteristic (ROC)')
plt.legend(loc='lower right')
plt.show()
```